FUTURE TECHNOLOGY AND ENGINEERING TEACHER

OVERVIEW

As the need for student proficiency in technology (as one area of STEM) is increasing, so is the need for qualified technology education teachers. Technology is advancing at a rapid rate and teachers are expected to adopt this technology as fast as it's developed. Applying leadership and 21st century skills, participants create a portfolio demonstrating their interest and aptitude for the teaching profession, which will include a lesson plan and accompanying activity that incorporate standards of technological literacy. Semifinalists engage the judges with the proposed activity. (Use the International Technology and Engineering Educators Association ITEEA website at www.iteea.org for more information about the technology standards.) Topics also should reflect Science, Technology, Engineering, and Mathematics (STEM) initiatives and integration. Lesson plans/activities that explore knowledge, creativity, and skills in the following areas are suggested:

- Medical technology
- Agricultural and biotechnology
- Power and energy technology
- Information and communication technology
- Transportation technology
- Manufacturing technology
- Construction technology

ELIGIBILITY

Two (2) individuals per chapter may participate.

TIME LIMITS

PRE-CONFERENCE/PRELIMINARY ROUND

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

C. All components of the chapter's entry (documentation portfolio PDF) must be finished, submitted, and accessible via the Internet by 11:59pm ET on a designated date mid-May.

SEMIFINAL ROUND

A. Semifinalists have fifteen (15) minutes for the presentation (and responses to judge questions) of their activity with the timing broken down as follows:

1. Three (3) minutes for any required set-up
2. Seven (7) minutes or less for the activity presentation
3. Three (3) minutes to answer questions from the judges
4. Two (2) minutes for breakdown and removal of materials.

B. One (1) point will be deducted for each ten (10)-second interval over the allotted time for the semifinal presentation.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE/PRELIMINARY ROUND

A. In preparation for this event, participants thoroughly research and select two (2) educational technologies or applications that can be integrated into the classroom.

B. Participants assemble a portfolio containing their research on educational technologies/applications, a lesson plan incorporating ITEEA standards, and an activity that accompanies their lesson.

C. Participants submit their portfolio by 11:59pm ET on a designated date in mid-May.

D. Submission information is provided on the TSA website under *Competition Updates*.

E. A list of the twelve (12) semifinalists (in random order) will be posted on the first full day of the conference.

SEMIFINAL ROUND

A. In preparation for the semifinal round, participants prepare a seven (7) minute presentation demonstrating their activity.

B. Participants report at the time and place stated in the conference program to sign up for a scheduled time to present the activity.

C. Participants report at the assigned time and place for the presentation.

D. Participants respond to questions pertaining to their entry.

E. Participants have fifteen (15) minutes for the presentation (and responses to judge questions) broken down as follows:

1. Three (3) minutes or less to set-up/prepare materials for the activity.
2. Seven (7) minutes or less to present the activity.
3. Three (3) minutes to answer questions from the judges.
4. Two (2) minutes to breakdown and remove materials.

F. The top ten (10) finalists are announced during the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, presentation, and/or communication pertaining to the entry.

PRE-CONFERENCE/PRELIMINARY ROUND

A. Documentation materials (comprising "a portfolio") are required and should be compiled into a single PDF file for submission. The portfolio must include the following pages, in this order:

1. Title page with the event title, the conference city and state, the year, and the participant's identification number; one (1) page
2. Table of contents; pages as needed
3. Research summaries; one (1) page maximum for each technology/application
4. Lesson plan/activity; pages as needed
5. Differentiation and Adaptation Plan; maximum of three (3) pages
6. Copies of handouts; pages as needed
7. References and resources; pages as needed
8. Student Copyright Checklist (see 2027 & 2028 Forms Appendix); one (1) page
9. Consent and Release Forms – Recognizable individuals pictured in any images (minors require parental consent) must give their written consent before the images can be used in this event (see 2027 & 2028 Forms Appendix); pages as needed
10. Resources and AI Reflection (see 2027 & 2028 Forms Appendix); maximum of four (4) pages

B. Research Summary

1. Using no more than two (2) pages, one (1) side only, participants summarize the information derived from researching each of the two (2) educational technologies or applications selected.
2. Participants should take into consideration the cost, intended age, and feasibility of the technologies/applications.
3. Selected technologies/applications may be specific to the lesson chosen or broadly applicable to the classroom. Physical technologies and digital applications are both permitted.

C. Lesson Plan

1. A lesson plan describing the rationale, goals and objectives, standards correlation, and a description of the lesson and activity—including assessment—must be submitted.
2. Lesson plan must be clearly labeled with the grade level for which it is intended.
3. ITEEA Standards for Technological Literacy must be used. Available online at www.iteea.org/stel

D. Differentiation and Adaptation Plan
   1. A differentiation and adaptation plan that reflects on how to modify the lesson for students with different ability levels while maintaining the same learning goals and standards must be submitted.
   2. Participants should include appropriate modifications for each component submitted (e.g. handouts, activities, assessments).
   3. Participants may choose to address physical, psychosocial, and/or cognitive disabilities in their approach.

E. Research summaries, essay, and lesson plan/activity must be the result of the participant's own efforts and not purchased or open source material. General content from either may be used, but it must be cited in MLA format.

F. Font size for any documentation component should be no less than size ten (10).

SEMIFNAL ROUND

A. Activity Presentation
   1. Participants do not teach the entire lesson plan. The semifinal round focus is the activity presentation.
   2. Participants design and create a presentation to walk judges through the proposed activity as if they were fellow educators being briefed on how to incorporate the activity into their classroom and how it meets educational standards. Participants may also choose to demonstrate the functionality of modifications made to the activity to meet different ability levels in the classroom.
   3. Participants describe the rationale, goals and objectives, standards correlation, and a description of the lesson and activity.
   4. ITEEA Standards for Technological Literacy must be used.
   5. Activity presentation must acknowledge the grade level for which it is intended.
   6. Activity Presentation timing:
      a. Presentation cannot exceed seven (7) minutes.
      b. There is no minimum length restriction.
      c. Activity presentation is timed from the first sound or voice to the final sound or voice.
      d. Presentation must include any applications being used and cite the applications in the credits or the introduction.

B. Entries must be the result of the participant's own efforts and not purchased or open source material.

C. Audio-visual materials such as charts, graphs, posters, displays, flip charts, transparencies, and models may be used.

D. Any audio visual equipment required for the presentation must be provided by the participant, including power strip with surge protector an extension cord.

E. Participants are not allowed to watch or hear the presentations of other participants.

F. Participants are encouraged to interact with the judges, who will act as students in the classroom.

G. One (1) point will be deducted for each ten (10)-second interval over the allotted time for the presentation, set up, and/or clean-up.

EVALUATION

PRELIMINARY ROUND

A. Documentation portfolio

SEMIFINAL ROUND

A. Presentation/interview

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Technology education teacher
- STEM teacher
- University professor
- Professional development trainer

Participant/Team ID# _____

FUTURE TECHNOLOGY AND ENGINEERING TEACHER

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Portfolio PDF was submitted pre-conference
☐ ENTRY NOT EVALUATED

PRELIMINARY DOCUMENTATION PORTFOLIO (90 points)				
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Portfolio components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Research Summaries (X1)	Only one (1) educational technology/ application is identified; summary is inadequate.	Two (2) educational technologies/ applications are adequately summarized.	Two (2) educational technologies/ applications are documented, with excellent summaries.	
Lesson Plan Components (X2)	Lesson plan is missing several of the stated components, including the standards correlation and/or instructional design focus; it is not creative or unique.	Lesson plan includes all of the stated components, and it is adequately organized, with an instructional design focus; it has some unique and creative aspects.	Lesson plan includes all of the stated components; it is well organized and has an instructional design focus; it is creative and unique.	
Handouts, Assessments, and Activity (X1)	Additional materials are poorly developed and do not have an impact on the lesson.	Additional materials are relatively well developed and have some impact on the lesson.	Additional materials are effective and greatly impact the lesson.	
Differentiation and Adaptation Plan (X1)	Differentiation and adaptation plan lacks thoughtfulness and feasibility; minimal attempts are made to adjust the submitted materials for varying ability levels.	Differentiation and adaptation plan demonstrates some thoughtfulness and a generally feasible approach; adequate attempts are made to adjust submitted materials for varying ability levels.	Differentiation and adaptation plan demonstrates extraordinary thoughtfulness and feasibility; exemplary attempts are made to adjust submitted materials for varying ability levels.	
Creativity (X1)	Submitted lesson plan demonstrates little creativity; additional materials lack unique or original perspectives of the chosen topic.	Submitted lesson plan demonstrates adequate creativity; additional materials convey some unique or original perspectives of the chosen topic.	Submitted lesson plan demonstrates exceptional creativity; additional materials frame the topic in a unique and original way.	
Resources and References (X1)	A minor attempt is made to document resources and references.	Several resources and references are documented.	There is clear evidence of many applicable resources and references.	

PRELIMINARY DOCUMENTATION PORTFOLIO (90 points) – continued				
Resources and AI Reflection (X1)	Form is incomplete or vague and does not clearly explain whether or how AI was used, showing little connection to the submission or understanding of effectiveness and ethics.	Form is complete and clearly explains AI use or the lack thereof with connections to the submission or theme/topic (when applicable), addressing effectiveness and ethical considerations with adequate analysis.	Form is complete, concise, and specific, critically evaluating how AI or non-AI tools shaped decisions and outcomes, with strong alignment to the submission or theme/topic (when applicable) and ethical reasoning.	
			PRELIMINARY DOCUMENTATION PORTFOLIIO SUBTOTAL (90 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (90 points)	

SEMIFINAL ACTIVITY PRESENTATION/INTERVIEW (100 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Presentation lacks organization; it is difficult to follow or understand.	Presentation is somewhat organized.	Presentation is clearly organized and easy to follow; it flows smoothly to the conclusion.	
Activity Components (X2)	Activity is missing several of the stated components, including the standards correlation and/or instructional design focus; it is not creative or unique.	Activity includes all of the stated components, and it is adequately organized, with an instructional design focus; it has some unique and creative aspects.	Activity includes all of the stated components; it is well organized and has an instructional design focus; it is creative and unique.	
Introduction (X1)	Introduction is weak, with little effort made to emphasize the topic and/or to generate interest and enthusiasm for the topic.	Sufficient effort is evident in the introduction, which creates some level of interest.	Introduction is effective, stimulating, and creates interest on the part of observers.	
Instructional Competence (X2)	Presenter's delivery of content lacks confidence; leadership and/or 21st century skills are not evident.	Presenter's delivery of content is generally professional and enthusiastic; leadership and/or 21st century skills are somewhat evident.	Presenter's delivery of content is professional, enthusiastic, confident, and full of personality; leadership and/or 21st century skills are clearly evident.	
Voice/Language (X1)	Participant conveys an inconsistent use of proper grammar, word pronunciation, and acceptable pitch and tone.	Participant generally uses proper grammar and pronunciation, and varies the use of tone and pitch.	Participant uses smooth and effective articulation, proper grammar, correct pronunciation, and varied tone and pitch throughout the presentation.	
Innovation/ Creativity (X2)	Presentation fails to convey innovation or originality.	Presentation is somewhat original and innovative in its delivery and topic development.	Presentation is imaginative and innovative in its delivery and topic development.	
Knowledge (X1)	Minimal knowledge of the subject is evident in the presentation and interview; content does not relate to the topic, and/or the participant does not convey an understanding of the topic.	Knowledge of the subject is evident during the presentation and interview, and the presenter relates and conveys a somewhat clear understanding of the topic.	Complete knowledge and understanding of the subject and relationship to the topic are conveyed throughout the lesson and interview.	

SEMIFINAL ACTIVITY PRESENTATION/INTERVIEW SUBTOTAL (100 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	

TIME DEDUCTIONS		
One (1) point will be deducted for each ten (10)-second interval over the allotted time for the semifinal presentation.		
Total time		
Presentation deduction		
TOTAL TIME DEDUCTION		

SEMIFINAL SUBTOTAL (100 points)	

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (190 points)	

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

FUTURE TECHNOLOGY AND ENGINEERING TEACHER EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator

B. Judges:

   1. Preliminary round, two (2) or more
   2. Semifinal round, two (2) or more (preferably the same judges as the preliminary round)

C. Assistants, two (2)

MATERIALS

A. Coordinator's packet, containing:

   1. Event guidelines, one (1) copy for the coordinator and for each judge
   2. TSA Event Coordinator Report

B. Stopwatch

C. Interview questions

D. Tables and chairs for participants and judges

E. Access to an AC electrical outlet

F. Copy of ITEEA publication *Standards for Technological Literacy*

RESPONSIBILITIES

PRE-CONFERENCE

A. National TSA will collect entries until 11:59pm ET on a designated date in mid-May and send out receipt confirmations to participants. The results are shared with the CRC manager, event coordinator, and assigned judges.

B. Review entries as they are submitted to the designated online storage utility.

C. Manage communication and Pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with National TSA and/or the Judge Manager.

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.

B. Report to the CRC room and check the contents of the coordinator's packet.

C. Review the event guidelines and check to see that enough personnel have been scheduled.

D. Inspect the area or room in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

E. At least one (1) hour before the event is scheduled to begin, meet with judges to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.

F. Ensure the judges have access to the online judging system.

PRELIMINARY ROUND

A. On the first full day of competition, post a list of the twelve (12) semifinalists in random order.

SEMIFINAL ROUND

A. Participants report at the time and place stated in the conference program to sign up for a presentation/interview time.

B. Participants report at the assigned time to the place stated, with supplies/materials (if applicable), for the presentation/interview.

C. Manage completion of the on-site activity presentation and interviews.

D. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and the CRC manager to determine either:

   1. To deduct twenty percent (20%) of the total possible points in this round
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

E. If necessary, manage security and the removal of materials from the event area.